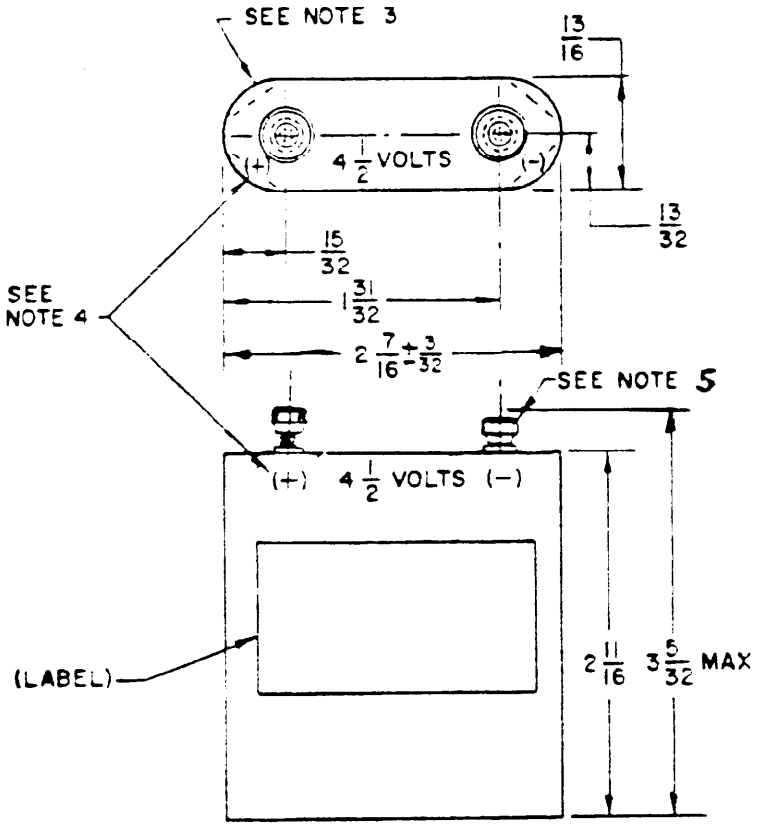
MIL-B-18/10D
21 May 1985
SUPERSEDING
MIL-B-18/10C
21 August 1976

MILITARY SPECIFICATION SHEET

BATTERY, DRY, BA-31

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dry battery described herein shall consist of this document and the latest issue of Specification MIL-B-18.



INCHES	MM
1/16	1.59
3/32	2.38
13/32	10.32
15/32	11.91
13/16	20.64
1 31/32	50.01
2 7/16	61.91
2 11/16	68.26
3 5/32	80.17

(D) denotes changes

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FSC-6135

NOTES:

1. All dimensions shown on figure are in inches. Unless otherwise specified, tolerance is ±1/16.
2. Metric equivalents specified in table are for general information only and are based upon 1 inch = 25.4 millimeters.
3. Diagonal or round corners are optional.
4. Voltage and polarity markings may be placed either on the top or front face of the battery.
5. Provide insulated knurled nuts with brass inserts.

REQUIREMENTS:

Dimensions and configuration: See figure.

Nominal voltage: 4.5 volts.

Usual number and type of cells: 3 "B" cells

Usual cell connection: Series

Terminals: stud and nut.

Weight (Maximum): 6 ounces.

(D) Capacity tests: When the battery is tested in accordance with the methods of examination and test of this Specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

(D) First Article inspection:

Visual and mechanical (External).
Battery voltage.
Vibration test.
Mechanical-shock test per MIL-STD-202, method 213, test condition I.
Insulation resistance test per MIL-STD-202, method 302, test condition Letter B, tolerance ±20 volts.
Capacity, D (without storage).
Jacket integrity test.

(D) Quality conformance inspection:

Visual and mechanical (External).
Battery voltage.
Insulation-resistance test per MIL-STD-202, method 302, test condition Letter B, tolerance ±20 volts.
Vibration test.
Mechanical-shock test per MIL-STD-202, method 213, test condition I.
Jacket integrity test.
Electrode leakage.
Capacity, D & T.

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Methods of examination and test:

Capacity tests: See requirements for capacity specified herein.

Storage -	Test	Period
	D	12 months
	T	90 days

Open-circuit voltage - After completion of storage for D test specified, the open circuit voltage shall be not less than 4.35 volts when measured at ambient storage conditions.

Discharge - The battery shall be discharged through 20 ohms for 4 minutes per hour, 10 hours per day, 5 days per week. This cycle shall be repeated continuously to a test-end voltage of 2.8 volts.

Closed circuit voltage - Use TS-183( )/U test per special marking or, with minimum permissible voltage as specified for TS-183( )/U, use a load resistance value of 19.95 ohms.

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

USE JACK NO.	MINIMUM PERMISSIBLE VOLTAGE
7	3.84

Unless expressly authorized, this test information shall apply to Army and Air Force applications only.

Custodian:
Army - ER
Navy - SH
Air Force - 99

Preparing activity:
Army - ER

Review activities:
Army- MI

Project Number: (6135-0155-7)

User activities:
Army - ME
Navy - YD, MC

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